# Jadd Akkad

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## Education

**University of Toronto** Toronto, ON
***Bachelor of Science in Computer Science, Mathematics, and Statistics*** *Present*
**Relevant Coursework:** Data Structures & Algorithms, Systems Programming, Operating Systems, Databases, Software Design, Software Engineering, Information Security, Computer Organization, Computational Complexity, Linear Algebra II, Statistics II, Intro to AI

## Professional Experience

**Nirvo** Jan. 2026 – Apr. 2026
***Fullstack Developer Intern***

- Led development of the **admin dashboard** using **TypeScript, Next.js, and React.js**, centralizing user management, session administration, and analytics.
- Developed **17 FastAPI routes** supporting account administration, session moderation, and analytics.
- **Reduced admin-list rendering time by up to 97%** through pagination, validated using controlled browser tests with 1,000 synthetic records.
- **Reduced database lookups by 99% (100 to 1)** through batch data retrieval, while preserving data accuracy.
- **Eliminated redundant API requests across 5 administrative actions** by updating frontend state without refetching entire user and session lists.
- **Improved usability across screen sizes** by refining responsive layouts, typography, and dashboard interactions.

## Software Engineering Projects (jaddboi.com)

**Gambot – Marvel Rivals Discord Bot** | *JavaScript, Node.js, discord.js, RivalsIsland API* Oct. 2026 – Present

- Developed a **Marvel Rivals Discord bot serving 358 daily active users**, providing player statistics and match history through interactive slash commands.
- **Enabled season and mode-specific performance tracking** by integrating the **RivalsIsland API** to retrieve player statistics and recent match results.
- **Simplified navigation of player statistics and match history** through paginated embeds, interactive scoreboards, and lazy loading.
- **Reduced redundant API requests** through in-memory caching while improving reliability with rate-limit handling, command cooldowns, and automated tests.

**Connect Four AI Agent** | *Python, Game Theory, Heuristics* Oct. 2026

- Built an adversarial agent using **Minimax** with **Alpha-Beta Pruning** for a 6x7 board.
- Designed a heuristic evaluation function that prioritized strategic positioning and won **over 80% of simulated matches**.
- Added **dynamic move ordering** to increase pruning and reduce computation time.
- Tuned **recursion depth** to balance decision quality with real-time performance.

**StudyFlow – AI Study Platform (In Progress)** | *Next.js, Supabase, Gemini API* Jul. 2026 – Present

- **Centralized academic management** by developing a full-stack platform for organizing courses, assignments, calendars, and grades using **Next.js and Supabase**.
- **Enabled semantic search across uploaded course materials** by implementing PDF extraction, text chunking, and vector embeddings.
- **Automated the generation of summaries and structured study notes** by integrating the **Gemini API** with course materials.
- **Protected student data** through authentication, row-level security, private storage, and ownership validation.
- **Improved background processing reliability** through atomic job claiming, persistent job states, retries, and fallback handling, supported by **51 automated tests**.

**Coding Quest Competition** | *Competitive Programming* Oct. 2022

- **Placed 1st overall** representing the International School of Kuala Lumpur in a competitive programming competition.
- **Achieved multiple top-10, top-20, and top-30 global problem solves**, demonstrating algorithmic problem-solving under competition constraints.

## Technical Skills

**Languages:** Python, Java, C, C#, SQL, JavaScript, TypeScript, HTML/CSS, Bash, R
**Frameworks & Libraries:** Next.js, React.js, Express.js, FastAPI, LangChain, LangGraph, Tailwind CSS, Socket.IO, discord.js
**Databases & Backend:** PostgreSQL, MongoDB, Mongoose, Supabase, REST APIs
**Tools & Systems:** Git/GitHub, Docker, Linux/Unix, Windows, macOS, Cursor, VS Code, IntelliJ, Microsoft Office, Google Workspace, Figma, Canva
**Core Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, MVC Architecture, Unit Testing, Agile/Scrum Development, Prompt Engineering
**Hobbies:** Player and coach on the UofT Marvel Rivals main team (backline in-game leader); former professional Brawlhalla player (ranked #2 in Asia); basketball, rock climbing, chess, violin, and more.